Introducing: Confidential Bare Metal



Three modes, three different adversaries

Each mode moves the trust boundary to match who operates the infrastructure

CBM — Confidential Bare Metal

TENANT RECEIVES
Whole host plus the NVLink fabric connecting them

PRIMARY ADVERSARY
Infrastructure owner

WHO RUNS THE HOST OS
Tenant

PRIMARY HW PROTECTION
DRTM measured launch
GPU in confidential mode

MAIN RESIDUAL EXPOSURE
CPU–GPU link on current platforms
Physical security

CVM — Confidential VM

TENANT RECEIVES
A VM on a shared host

PRIMARY ADVERSARY
Privileged host software and co-resident tenants

WHO RUNS THE HOST OS
Infrastructure Owner

PRIMARY HW PROTECTION
CPU TEE per VM
GPU in confidential mode

MAIN RESIDUAL EXPOSURE
Physical security

CoCo — Confidential Containers

TENANT RECEIVES
Kubernetes pods, each in its own confidential VM

PRIMARY ADVERSARY
Kubernetes control plane

WHO RUNS THE HOST OS
Infrastructure Owner

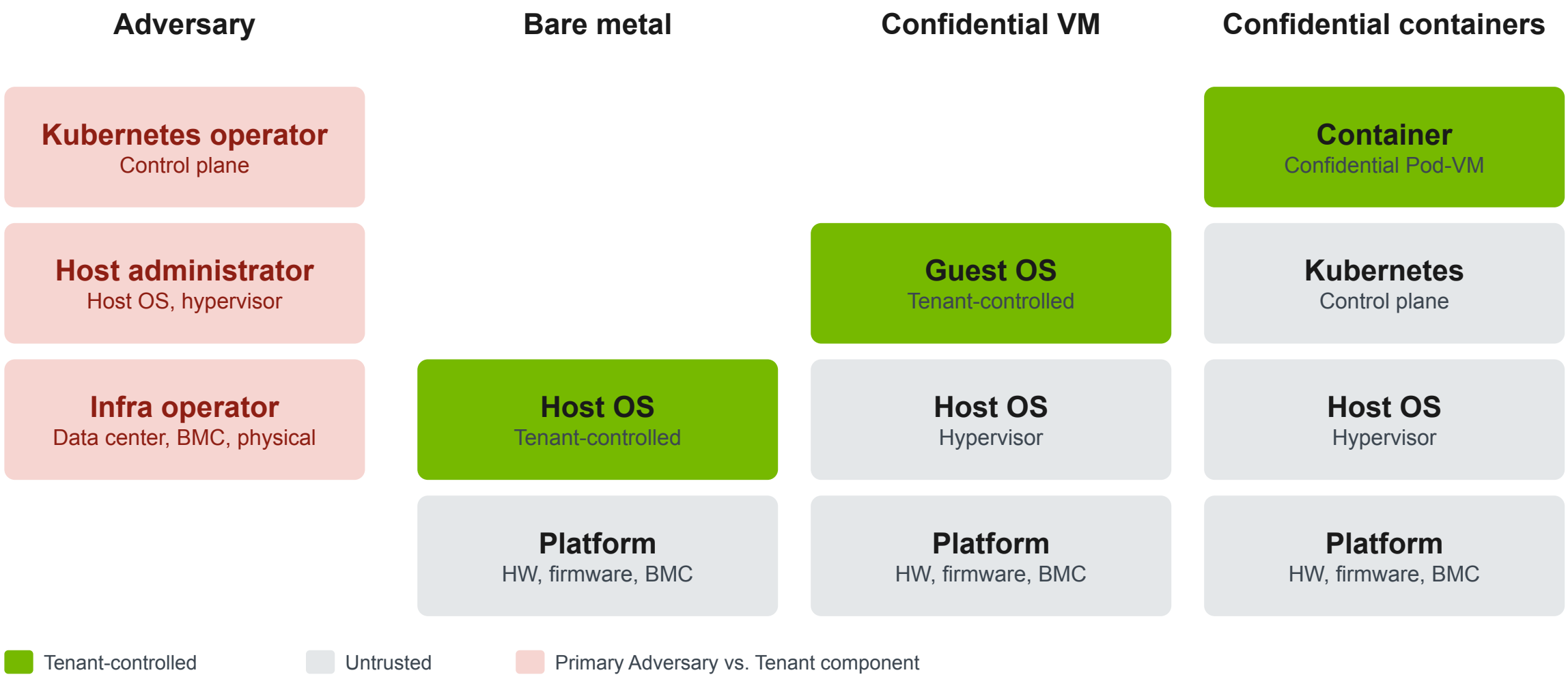
CPU-SIDE PROTECTION
CPU TEE per pod
GPU in confidential mode

MAIN RESIDUAL EXPOSURE
Control-plane-supplied config, volumes
Physical security

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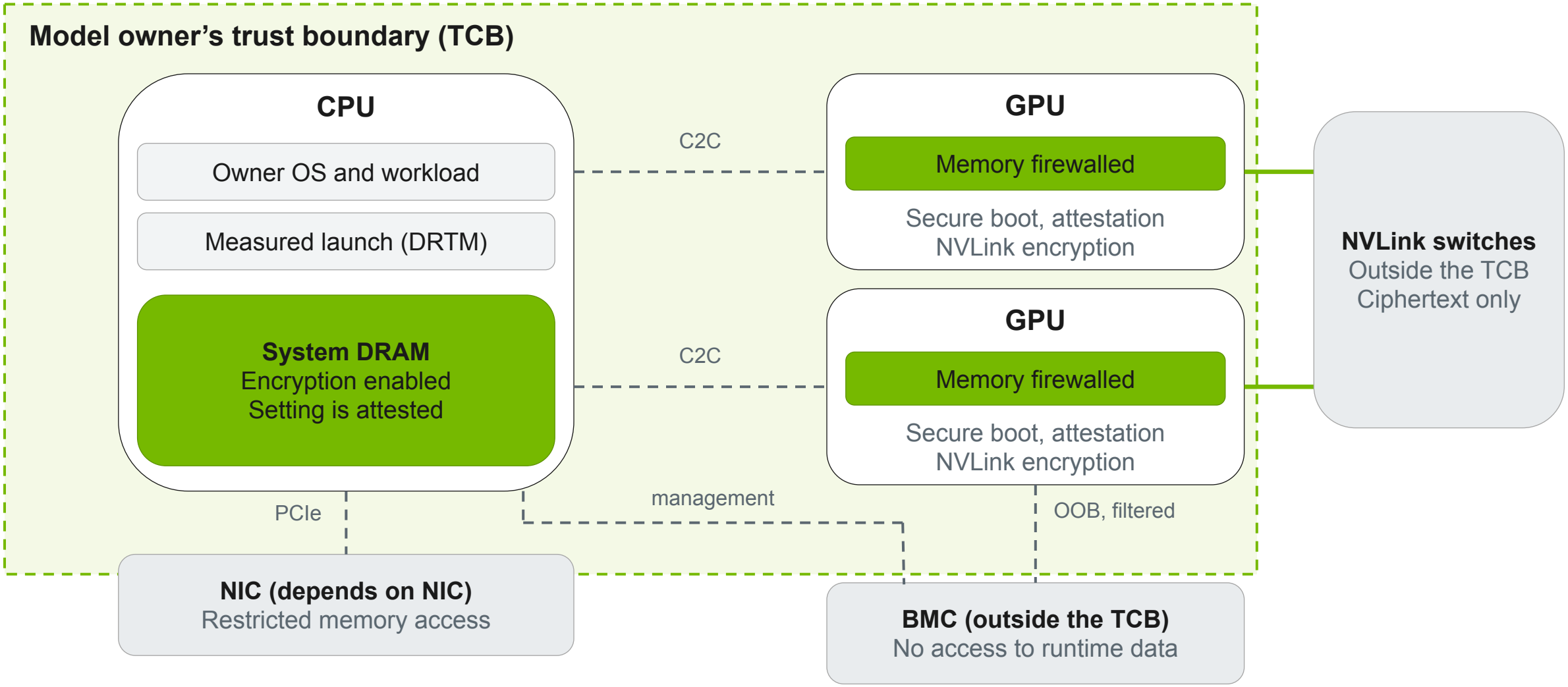
Confidential Computing Modalities

Each mode moves the tenant's layer higher; everything beneath it is untrusted



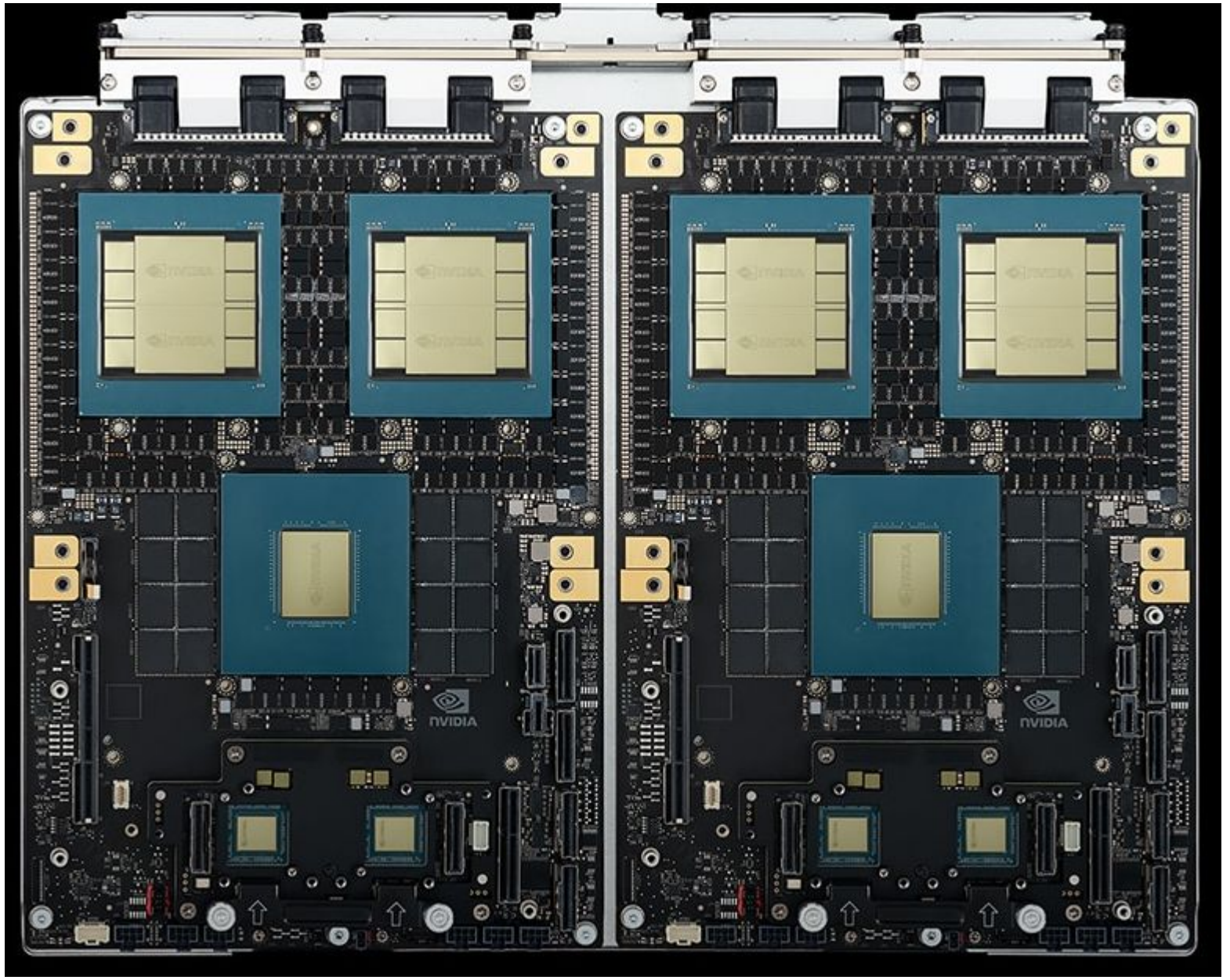
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CBM trust boundary



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Host Measurement - Dynamic Root of Trust for Measurement

Operator boot software drops out of the chain, while NVIDIA's secure firmware stays attested from boot

Boot-time evidence from the security controller

NVIDIA boot chain | Secure firmware | Fuses, lifecycle

Not in the new trust chain

UEFI / BIOS | EFI drivers | Option ROMs | Boot loaders

DL Event: the dynamic root of trust starts here

Other cores off, DMA blocked, interrupts masked, fresh measurement chain

D-CRTM

Trust anchor

- Launch firmware (DCE)
- Firmware signing key
- Debug, trace state
- Lifecycle state

Launch firmware

Called DCE in the spec

- Kernel image (DLME)
- Kernel entry point
- Kernel key, version
- End marker

Kernel

Called DLME in the spec

- Event log
- Trusted memory map
- ACPI tables, hashes

Measured at launch | Not in trust chain | Handed to the kernel | Launch event | Boot-time evidence

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